Department of Economics
Columbia University

UN3412
Spring 2026

Problem Set 5
Introduction to Econometrics
for Section 1
Erden

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1. (30p) Some U.S. states have enacted laws that allow citizens to carry concealed weapons. These laws are known as "shall-issue" laws because they instruct local authorities to issue a concealed weapons permit to all applicants who are citizens, mentally competent, and have not been convicted of a felony (some states have some additional restrictions). Proponents argue that, if more people carry concealed weapons, crime will decline because criminals are deterred. Opponents argue that crime will increase because of accidental or spontaneous use of the weapon. In this exercise, you will analyze the effect of concealed weapons laws on three different categories of crimes: violent crimes; robberies (such as the robbery of a convenience store); and murder (many of which are spontaneous acts of passion).

Data file and the description of variables are at the end of this problem set.
Estimate
(i) (3p) a regression of ln(vio) against shall and
(ii) (3p) a regression of ln(vio) against shall, incarc_rate, density, avginc, pop, pb1064, pw1064, and pm1029.

(a) (8p) Interpret the coefficient on shall in regression (ii). Is this estimate large or small in a "real-world" sense?

(b) (8p) Does adding the control variables in regression (ii) change your conclusions about the effect of a shall-carry law, relative to regression (i)?

(c) (8p) Suggest a variable which varies across states but plausibly varies little, or not at all, over time, and which plausibly could cause omitted variable bias in regression (ii).

2. (30p) **Read the notes to Tables 1–3**, then fill in the tables.

Table 1
The Effect of Concealed Handgun Laws on Violent Crime: Regression Results

Dependent variable: ln(*vio*)

	(1)	**(2)**	**(3)**	**(4)**	**(5)**
Coefficient on *shall*	( )	( )	( )	( )	( )
State characteristic control variables[a]?	no	yes	yes	yes	yes
State fixed effects?	no	no	yes	yes	yes
Year fixed effects?	no	no	no	yes	yes
F-statistic testing the hypothesis that the state fixed effects are zero	–	–	( )	( )	–
F-statistic testing the hypothesis that the year fixed effects are zero	–	–	–	( )	( )
HAC (clustered) SEs?	no	no	no	no	yes
n					

Notes: All regressions include an intercept. For regressions (1) – (4), heteroskedasticity-robust standard errors appear in parentheses below estimated coefficients; for regression (5), the standard errors are heteroskedasticity-robust and clustered at the state level, so as to allow for serial correlation in the error within a state. *p*-values appear in parentheses beneath heteroskedasticity-robust *F*-statistics (or, for regression (5), heteroskedasticity-robust-clustered *F*-statistic).
[a]Regressions with "state characteristic control variables" include the following regressors: *incarc_rate, density*, *avginc*, *pop*, *pb1064*, *pw1064*, *pm1029*.

Table 2
The Effect of Concealed Handgun Laws on Robberies: Regression results
Dependent variable: ln(*rob*)

	(1)	**(2)**	**(3)**	**(4)**	**(5)**
Coefficient on *shall*	( )	( )	( )	( )	( )
State characteristic control variables[a]?	no	yes	yes	yes	yes
State fixed effects?	no	no	yes	yes	yes
Year fixed effects?	no	no	no	yes	yes
F-statistic testing the hypothesis that the state fixed effects are zero	–	–	( )	( )	–
F-statistic testing the hypothesis that the year fixed effects are zero	–	–	–	( )	( )
HAC (clustered) SEs?	no	no	no	no	yes
n					

Notes: See the notes to Table 1.

Table 3
The Effect of Concealed Handgun Laws on Murders: Regression results
Dependent variable: ln(*mur*)

	(1)	**(2)**	**(3)**	**(4)**	**(5)**
Coefficient on *shall*	( )	( )	( )	( )	( )
State characteristic control variables[a]?	no	yes	yes	yes	yes
State fixed effects?	no	no	yes	yes	yes
Year fixed effects?	no	no	no	yes	yes
F-statistic testing the hypothesis that the state fixed effects are zero	–	–	( )	( )	–
F-statistic testing the hypothesis that the year fixed effects are zero	–	–	–	( )	( )
HAC (clustered) SEs?	no	no	no	no	yes
n	1173	1173	1173	1173	1173

Notes: See the notes to Table 1.

3. (40p) In order to determine the effects of stock market performance on individual firm's performance, you collect a daily frequency data on firm's performance for ***n*** sample of publicly listed companies for the period of January 1, 2021 – January 1, 2026 on a (trading) daily frequency.

(a) (8 p) What measures of firm performance would you include in an equation? What are some of the timing issues?
(b) (8 p) What other factors might you control for in the equation?
(c) (8 p) Write an equation that allows you to estimate the effects of the overall stock market performance on the percentage change in firm's stock price. How would you estimate this equation? Why would you choose this method?
(d) (16 p) Now, in order to implement your strategy discussed above, go to Yahoo Finance website and collect the stock price data of any ten U.S. companies of your choice as well as the stock price for S&P500 (i.e., as your measure of overall market performance and its ticker symbol is ^GSPC) for the above sample. Use the adjusted close price (column) as your price data for your ***n =10*** firms and compute the percentage change of these series as your performance measure. Make sure that you arrange your data as a panel data format by stacking these 10 companies' dataset. Then i) run a cross section fixed effect panel data model of firm performance on market performance only; and ii) explain (i.e., not estimate) how you would estimate a model or control for time fixed effects as well.

DATA DESCRIPTION, FILE: handguns.dta

This dataset is a balanced panel of data on 50 US states, plus the District of Columbia (for a total of 51 "states"), by year for 1977 – 1999, so an observation is a given state in a given year. There are a total of 51 states · 23 years = 1173 observations. Crime rates from the Bureau of Justice Statistics. The data set contains more variables than those listed below; brief definitions of those variables are given in the .dta file.

Variable	Definition
vio	violent crime rate (incidents per 100,000 members of the population)
rob	robbery rate (incidents per 100,000)
mur	murder rate (incidents per 100,000)
shall	= 1 if the state has a shall-carry law in effect in that year = 0 otherwise
incarc_rate	incarceration rate in the state in the previous year (sentenced prisoners per 100,000 residents; value for the previous year)
density	population per square mile of land area, divided by 1000
avginc	real per capita personal income in the state, in thousands of dollars
pop	state population, in millions of people
pm1029	percent of state population that is male, ages 10 to 29
pw1064	percent of state population that is white, ages 10 to 64
pb1064	percent of state population that is black, ages 10 to 64
stateid	ID number of states (Alabama = 1, Alaska = 2, etc.)

STATA HINTS

You will find it advantageous to use .do and .log files in this problem set. In addition, depending on your configuration, you may or may not need the first two commands below. STATA commands for fixed effects regression will be mentioned in class and discussed in section.

STATA command	Description
`set memory 50m`	Sets the memory used by STATA to 50m (do this prior to loading the data)
`set matsize 300`	Allows for large matrices
`describe`	lists the variables currently in memory and gives their definitions if available.
`tabulate state, generate(st)`	Creates a full set of indicator variables for the categories in the variable state: st1 equals one when state=1 and zero otherwise, st2 equals one when state=2 and zero otherwise, etc. (see the course STATA tutorial)
`testparm st*`	Computes the *F*-statistic testing the hypothesis that all the coefficients are zero on the variables having the first two letters st in the previous regression (see the course STATA tutorial)
`areg y x1 x2, absorb(state) r cluster(state)`	fixed effects regression of *y* on *x1* and *x2*, including state fixed effects and heteroskedasticity-robust clustered standard errors

<u>Following questions will not be graded, they are for you to practice and will be discussed at the recitation:</u>

1. [ungraded] SW Exercise 10.1
2. [ungraded] SW Exercise 10.5
3. [ungrade] SW Empirical Exercises 10.2